

# Morning Paper Report - Friday, June 26, 2026

Recent, non-repeated papers matched to visual working memory, visual search, modeling, working-memory representation, neural-network memory models, AI for human memory, and egocentric predictive-coding memory themes.

## 1. Perturbing human V1 degrades the fidelity of visual working memory

Mrugank Dake, Clayton E. Curtis  
bioRxiv, 2025

Source: Semantic Scholar; matched terms: visual working memory, working memory, fidelity

Link: <https://www.semanticscholar.org/paper/83ea4fbc8ff13c245a0054bab9a8432e077d30af>

**Goal:** Decades of macaque research established the importance of prefrontal cortex for working memory

**Method:** Moreover, concurrently measured electroencephalography confirmed that stimulation disrupted not only memory behavior, but neurophysiological signatures of working memory

**Results:** These results change the question from whether visual cortex is necessary for working memory to what mechanisms it uses to support memory

**Conclusion:** These results change the question from whether visual cortex is necessary for working memory to what mechanisms it uses to support memory

## 2. Investigating the effects of context, visual working memory, and inhibitory control in hybrid visual search

Alessandra Barbosa, G. Ruarte, Anthony J. Ries, J. Kamienkowski, Matias J. Ison  
Frontiers in Human Neuroscience, 2024

Source: Semantic Scholar; matched terms: visual working memory, visual search, working memory

Link: <https://www.semanticscholar.org/paper/e78c33d9f97cecb3cf0b9b4b928b140032fb3a96>

**Goal:** Introduction In real-life scenarios, individuals frequently engage in tasks that involve searching for one of the distinct items stored in memory

**Method:** In an online experiment, 110 participants searched for potential targets in images with and without context

**Results:** Results We show that, in target present trials, the main hallmarks of hybrid search remain present, with a linear relationship between reaction time and visual set size and a logarithmic relationship between reaction time and memory set size

**Conclusion:** Inhibitory control, when relationships were significant, could account for only a small portion of the variability in the data Discussion This study provides insights into the effects of context and individual differences on search efficiency and termination.

## 3. Wearable IoT and Artificial Intelligence Techniques for Leveraging the Human Activity Analysis

Lina Sheker, Vishwanath Petli, K. Satish Reddy  
Journal of Smart Internet of Things, 2024

Source: Semantic Scholar; matched terms: precision, modeling, neural network, neural networks, deep learning, artificial intelligence

Link: <https://www.semanticscholar.org/paper/218fadfff0b621943f8e7a03a8a33074786e8baf>

**Goal:** Abstract The integration of wearable Internet of Things (IoT) devices with advanced Artificial Intelligence (AI) techniques has significantly transformed human activity analysis (HAA), It enables precise, real-time monitoring of diverse physical behaviors

**Method:** Abstract The integration of wearable Internet of Things (IoT) devices with advanced Artificial Intelligence (AI) techniques has significantly transformed human activity analysis (HAA), It enables precise, real-time monitoring of diverse physical behaviors

**Results:** The dataset comprises accelerometer and gyroscope data collected from wearable devices during various activities, offering a robust foundation for HAR

**Conclusion:** Experimental results reveal the model's consistent and exceptional performance, highlighting its superiority over existing state-of-the-art approaches The findings underscore the practical viability of deploying such intelligent systems in real-world environments, representing a notable progression in the evolution of human activity analysis.

#### 4. Emotional content affects the fidelity of visual working memory recall.

Zeinab Haghian, A. Vahabie, Ehsan Rezayat  
Cognition & Emotion, 2025

Source: Semantic Scholar; matched terms: visual working memory, working memory, fidelity

Link: <https://www.semanticscholar.org/paper/b86c1c765c36262d18d9e72ec44364f82da869bf>

**Goal:** Our working memory (VWM) is susceptible to distortions influenced by various sources

**Method:** Thirty-one participants recalled the emotional valence of a target face, specified by its serial position (1, 2, or 3), by selecting a response from a continuous spectrum of 19 morphed faces

**Results:** Emotional content systematically distorts VWM representations, largely driven by a regression toward the mean This suggests that fundamental cognitive mechanisms, such as central tendency bias, are key drivers of how emotional information is maintained and recalled, with recall fidelity being shaped by an interplay between stimulus intensity, cognitive load, and temporal dynamics.

**Conclusion:** This suggests that fundamental cognitive mechanisms, such as central tendency bias, are key drivers of how emotional information is maintained and recalled, with recall fidelity being shaped by an interplay between stimulus intensity, cognitive load, and temporal dynamics.

#### 5. Investigating the effects of contextual information, visual working memory, and inhibitory control in hybrid visual search

Alessandra Barbosa, Anthony J. Ries, Juan E. Kamienkowski, Matias J. Ison  
2023

Source: Crossref; matched terms: visual working memory, visual search, working memory

Link: <https://doi.org/10.1101/2023.09.29.560143>

**Goal:** Abstract In real-life scenarios, individuals frequently engage in tasks that involve searching for one of various items stored in memory

**Method:** In an online experiment, 110 participants searched for potential targets in images with and without context

**Results:** We show that, in target present trials, the main hallmarks of hybrid search remain present, with a linear relationship between reaction time and visual set size, and a logarithmic relationship between reaction time and memory set size

**Conclusion:** In target-absent trials we found large differences between context-present and absent conditions, suggesting participants' adoption of adaptive strategies